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Project Briefing Elevator Controller System

2025/2026 - 2

FACULTY OF COMPUTING



Overview

- The project is to implement knowledge from this course by simulating an imaginary case study.
- The project is done in a group of 3 or 4 students
- Strategies used:
 - Group self explore
 - Creative problem solution and design
 - Good reporting and presentation



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Case Study

To design an electronic controller for elevator / lift in a hotel building





Case Study

- Scenario:
 - A hotel visitor will use the elevator to go up / down from one level to another
 - The visitor will key in the desired level to go, tag the visitor room card, close the elevator door and push UP / DOWN button
 - Then the elevator will go UP / DOWN by counting UP / DOWN one level to another until it reaches to desired level



Scenario

Lift UP

- Key in level up
- Tag in card hotel
- Door Close
- Lift goes up to destination
- Lift stop at destination
- Door Open

Lift DOWN

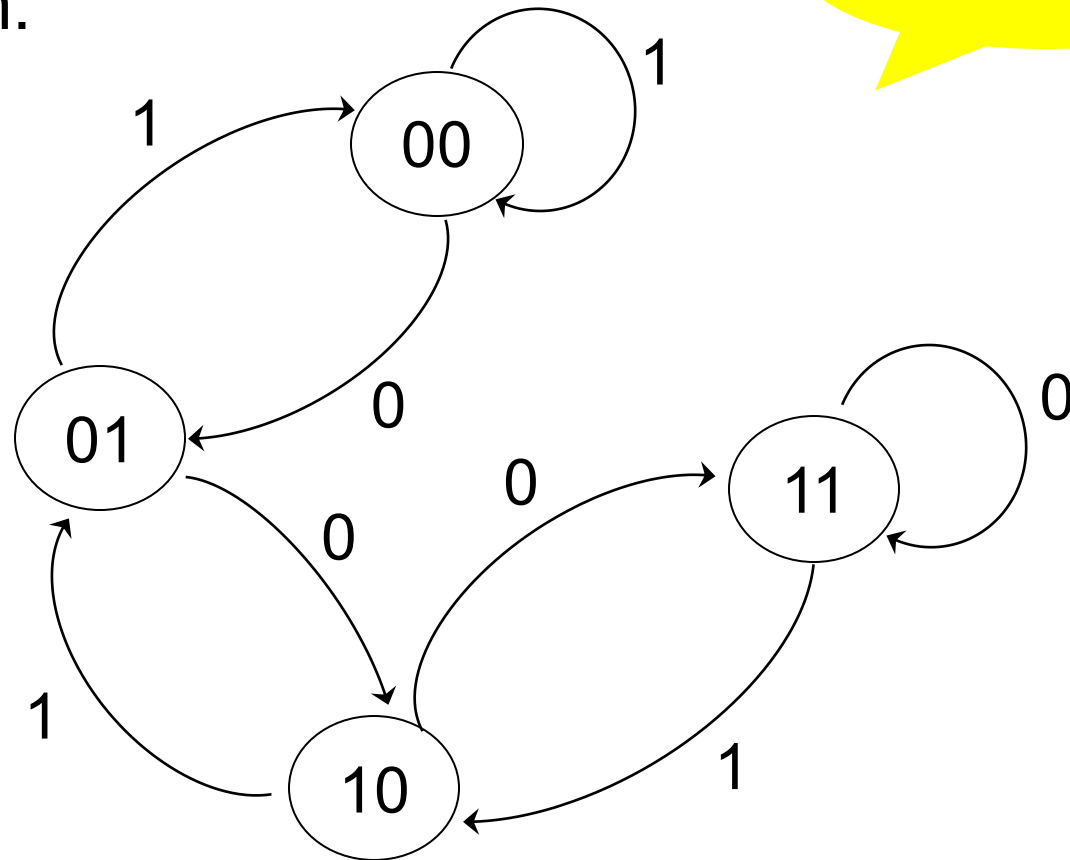
- Key in level down
- Tag in card hotel
- Door Close
- Lift goes down to destination
- Lift stop at destination
- Door Open



- 4 floor elevator system
 - For level 0, 1, 2, and 3
 - Use 2-bit up/down counter
 - 0 = count UP
 - 1 = count DOWN
 - Function in synchronous mode
 - Uses D flip flop

Basic Design

- State diagram:



Saturated

Basic Design

- Next State and flip flop transition table:

Input, X	Present State		Next State		D FF	
	$Q1_n$	$Q0_n$	$Q1_{n+1}$	$Q0_{n+1}$	D1	D0
0	0	0	0	1	0	1
0	0	1	1	0	1	0
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	1	0	1
1	1	1	1	0	1	0

Basic Design

- K-map:

Input, X	Present State		Next State		D FF	
	$Q1_n$	$Q0_n$	$Q1_{n+1}$	$Q0_{n+1}$	D1	D0
0	0	0	0	1	0	1
0	0	1	1	0	1	0
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	1	0	1
1	1	1	1	0	1	0

Q_0

XQ_1

	0	1
00	0	1
01	1	1
11	0	1
10	0	0

D_1

$$D_1 = \bar{X}Q_0 + \bar{X}Q_1 + Q_1Q_0$$

Q_0

XQ_1

	0	1
00	1	0
01	1	1
11	1	0
10	0	0

D_0

$$D_0 = \bar{X}\bar{Q}_0 + \bar{X}Q_1 + Q_1\bar{Q}_0$$

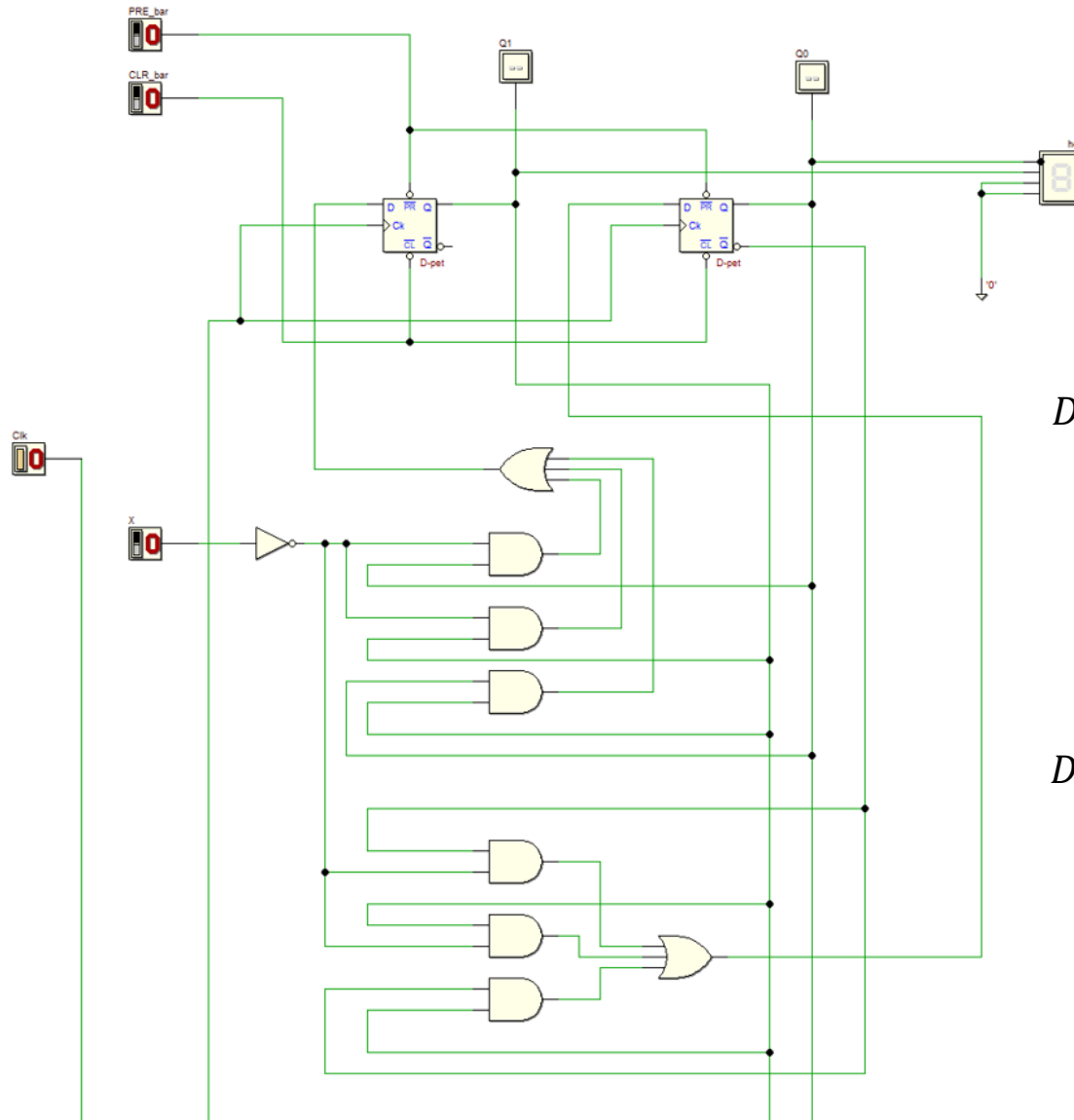


Basic Design in DEEDS

- Flip-flop using D Flip-flop
- **Comparator** for current state and targeted level floor using XNOR and NAND
- Pattern detector for **clock enabler** using AND gate.
 - If output AND gate is HIGH, clock is enabled, and counter can run.
- Use **push button** instead of clock generator.
 - This is to eliminate undesirable state during states transition

Design in DEEDS

2-bit UP / DOWN
counter using
D flip flops



$$D_1 = \bar{X}Q_0 + \bar{X}Q_1 + Q_1Q_0$$

$$D_0 = \bar{X}\bar{Q}_0 + \bar{X}Q_1 + Q_1\bar{Q}_0$$

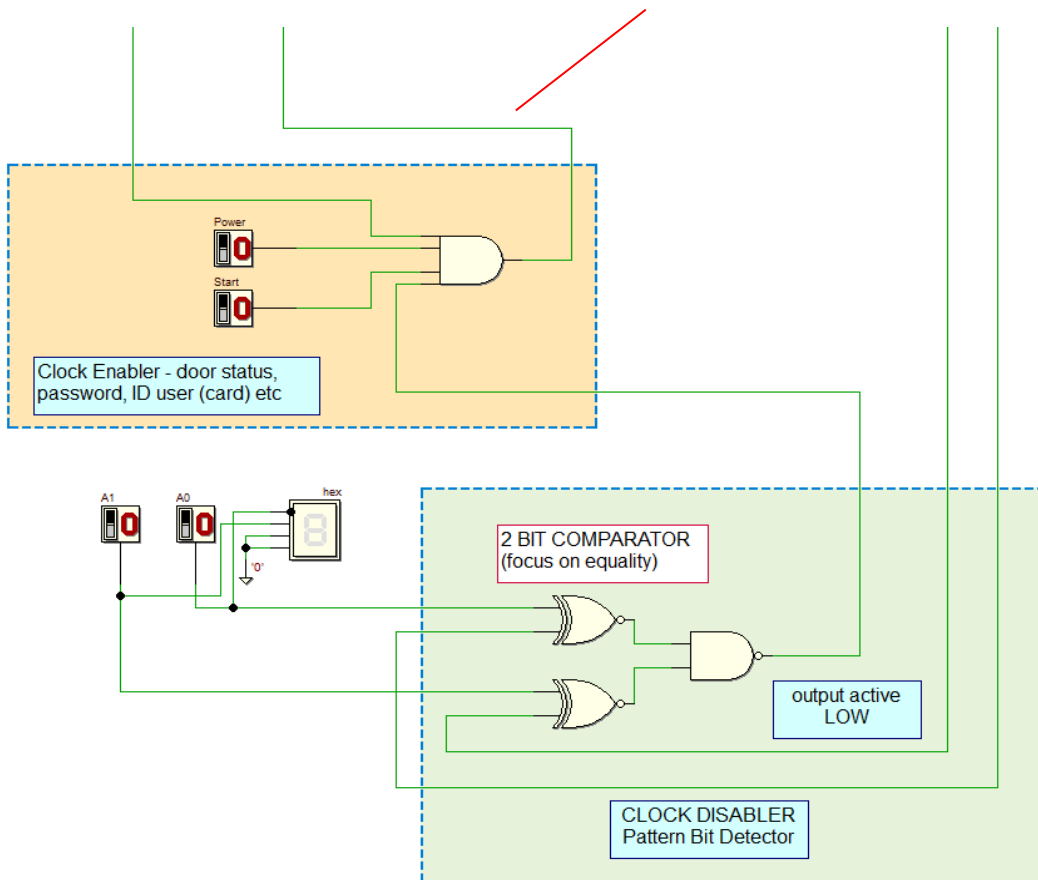
		Q ₀	
XQ ₁		0	1
	00	0	1
	01	1	1
	11	0	1
	10	0	0

		Q ₀	
XQ ₁		0	1
	00	1	0
	01	1	1
	11	1	0
	10	0	0

Design in DEEDS

Clock Enabler and Comparator

To clock input



Rule

1. Clock to the counter will be enabled when **HIGH** output goes to **clk** input at D flip-flop.
2. Gate **AND** will goes **HIGH** when CLK (push button) is **HIGH** and Power is **HIGH** and Start is **HIGH** and output comparator **IS NOT EQUAL**.
3. If comparator is **equal**, then NAND gate will generate output **LOW**, thus will **disable** the **clock** to D flip-flop.

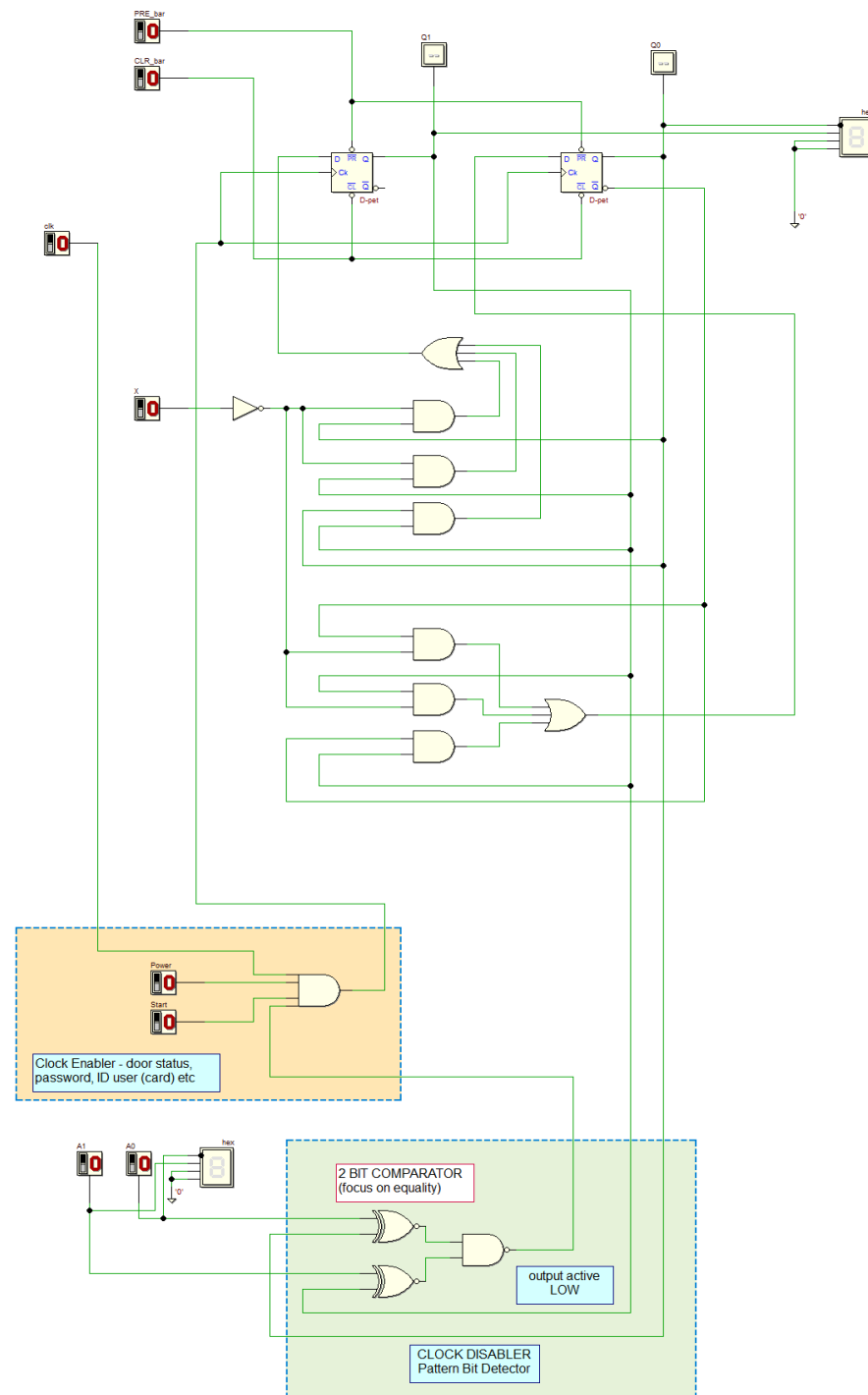
Operation:

Set value (UP)

- PRE_bar=CLR_bar= 1
- X=0 (count up)
- Power=1
- Give desired floor, e.g = 3 (from current floor = 0)
- Start=1
- Push clock button (Clk), counter will do count up from initial (current 0) to desired floor 3.

Set value (Down)

- PRE_bar=CLR_bar= 1
- X=1 (count down)
- Power=1
- Give desired floor, e.g = 0 (from current floor =3)
- Start=1
- Push clock button (Clk), counter will do count down from initial (current 3) to desired floor 0.





Your job ...

- You must enhance the previous design to **3-bit synchronous counter** (count UP/DOWN) to support 8 floors hotel elevator.
- Thus, must show **4 inputs K-map**, i.e., X , Q_2 , Q_1 , Q_0 to design the counter circuit and related states / flip flop transition tables.
- Use **JK flip flop** in your circuit design.



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Your job ...

- Introduce these features to the input controller:
 - to detect visitor card hotel
 - to indicate door open / close
 - to use passcode



PROJECT REPORT CONTENT

- Background / overview
- Problem Statement
- Suggested Solution
 - must include all the diagrams / tables and explain
 - ... state diagram, state table, transition table
- System Implementation (must include DEEDS drawing and explain)
- Conclusion and Reflection
 - Summarizes the whole project. Write your reflections.
 - Describe your achievements, strength and weaknesses.
 - Propose future work to improve your design



Report Format

Format

- Times New Roman 12 font size
- 1.5 spacing

What should be in the report?

- TITLE PAGE
- DEDICATION & ACKNOWLEDGEMENT
- TABLE OF CONTENTS
- REPORT CONTENT
- REFERENCES
- APPENDICES

Under Appendices:

- List of tasks distribution among members of group
- Photo(s) of your group working together
- And others supporting materials that will strengthen the explanation of your project.



- Prepare 15 minutes video to demo / demonstrate your project outcome
- Explain how you run the circuit simulation
- Produce the demo video in mp4 format
- Upload the demo video to Elearning



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Submission

- Project report
- Deeds source file (*.pbs)
- Demo video
- **DUE: 28 June 2026**
- Submit via elearnppsm.utmspace.edu.my